Agentic AI Assignment 5

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Model: meta-llama/Llama-3.2-1B-Instruct

Tasks: Summarization, Sentiment Analysis, Interview Questions, AI Tutor

Python Code

The program loads the Hugging Face token from a .env file, sends formatted prompts to the Featherless AI provider, and uses the Llama 3.2 1B Instruct model for all four tasks.

```
import os
from dotenv import load_dotenv

from huggingface_hub import InferenceClient
from langchain_core.prompts import PromptTemplate

# Load environment variables
load_dotenv()

token = os.getenv("HF_TOKEN")

print("Token loaded:", bool(token))

# Hugging Face client
client = InferenceClient(
    api_key=token,
    provider="featherless-ai"
)

# Model
model = "meta-llama/Llama-3.2-1B-Instruct"


def generate_response(prompt, max_tokens):
    response = client.chat.completions.create(
        model=model,
        messages=[
            {
                "role": "user",
                "content": prompt
            }
        ],
        max_tokens=max_tokens
    )

    return response.choices[0].message.content


while True:

    print("\n============================")
    print("       AI TASK MENU")
    print("============================")
    print("1. Text Summarization")
    print("2. Sentiment Analysis")
    print("3. Interview Questions")
    print("4. AI Tutor")
    print("5. Exit")

    choice = input("\nEnter your choice: ")

    # ----------------------------------
```

```
    # TASK 1: SUMMARIZATION
    # ----------------------------------
    if choice == "1":

        text = input("\nEnter text to summarize: ")

        prompt = PromptTemplate.from_template(
            "Summarize the following text in exactly 3 sentences:\n\n{text}"
        )

        final_prompt = prompt.invoke({"text": text})

        result = generate_response(
            final_prompt.to_string(),
            100
        )

        print("\nSummary:")
        print(result)

    # ----------------------------------
    # TASK 2: SENTIMENT ANALYSIS
    # ----------------------------------
    elif choice == "2":

        text = input("\nEnter a sentence or review: ")

        prompt = PromptTemplate.from_template(
            """Analyze the sentiment of the following text.

Return ONLY one of these three labels:
Positive
Negative
Neutral

Text:
{text}"""
        )

        final_prompt = prompt.invoke({"text": text})

        result = generate_response(
            final_prompt.to_string(),
            30
        )

        print("\nSentiment:")
        print(result)

    # ----------------------------------
    # TASK 3: INTERVIEW QUESTIONS
    # ----------------------------------
    elif choice == "3":

        topic = input("\nEnter interview topic: ")

        prompt = PromptTemplate.from_template(
```

```
            """Generate 5 interview questions about the following topic.

Topic: {topic}

Number the questions from 1 to 5.
Give only the questions."""
        )

        final_prompt = prompt.invoke({"topic": topic})

        result = generate_response(
            final_prompt.to_string(),
            150
        )

        print("\nInterview Questions:")
        print(result)

    # ----------------------------------
    # TASK 4: AI TUTOR
    # ----------------------------------
    elif choice == "4":

        topic = input("\nEnter topic: ")
        level = input("Enter student level (Beginner/Intermediate/Advanced): ")

        prompt = PromptTemplate.from_template(
            """Act as an AI tutor.

Teach the following topic to a student.

Topic: {topic}
Student Level: {level}

Explain the concept clearly and simply.
Give a small example.
End with one practice question."""
        )

        final_prompt = prompt.invoke({
            "topic": topic,
            "level": level
        })

        result = generate_response(
            final_prompt.to_string(),
            150
        )

        print("\nAI Tutor:")
        print(result)

    # ----------------------------------
    # EXIT
    # ----------------------------------
    elif choice == "5":
```

```
        print("\nExiting program...")
        break

    else:

        print("\nInvalid choice. Please enter 1-5.")
```

Execution Output

The following console output is reproduced from the provided execution, including the incomplete model responses where generation stopped at the configured token limit.

```
============================
       AI TASK MENU
============================
1. Text Summarization
2. Sentiment Analysis
3. Interview Questions
4. AI Tutor
5. Exit

Enter your choice: 1

Enter text to summarize: Artificial intelligence is rapidly changing many industries by
enabling computers to perform tasks that normally require human intelligence. AI
systems can analyze large amounts of data, understand language, recognize images, make
predictions, and assist people in making decisions. These technologies are being used
in healthcare, education, finance, transportation, and many other fields. However, the
increasing use of AI also creates challenges related to privacy, security, employment,
bias, and responsible use.

Summary:
Here is a 3-sentence summary of the text:

Artificial intelligence (AI) is rapidly changing various industries by automating tasks
that require human intelligence, such as data analysis, language understanding, and
image recognition. AI is being applied in healthcare, education, finance,
transportation, and other fields, but also raises concerns about privacy, security,
employment, and bias. As AI continues to evolve, it is essential to address these
challenges and ensure responsible use to mitigate potential negative impacts.

============================
       AI TASK MENU
============================
1. Text Summarization
2. Sentiment Analysis
3. Interview Questions
4. AI Tutor
5. Exit

Enter your choice: 2

Enter a sentence or review: I am extremely disappointed with this product. It stopped
working after only two days and the customer support was not helpful.

Sentiment:
Negative

============================
       AI TASK MENU
============================
1. Text Summarization
2. Sentiment Analysis
3. Interview Questions
```

```
4. AI Tutor
5. Exit

Enter your choice: 3

Enter interview topic: Python

Interview Questions:
1. What is the primary advantage of using the Python programming language, and how does
it differ from other languages in the industry?

2. Can you explain the concept of object-oriented programming (OOP) in Python, and
provide an example of how it is applied in a real-world scenario?

3. What are some of the key libraries and frameworks available for data science and
machine learning tasks in Python, and how do they enhance the development process?

4. How do you handle errors and exceptions in Python code, and what are some best
practices for debugging and troubleshooting?

5. What are some of the most common pitfalls and mistakes that beginners make when
learning Python, and how can they be avoided to ensure a smooth learning experience?

============================
       AI TASK MENU
============================
1. Text Summarization
2. Sentiment Analysis
3. Interview Questions
4. AI Tutor
5. Exit

Enter your choice: 4

Enter topic: Machine Learning
Enter student level (Beginner/Intermediate/Advanced): Beginner

AI Tutor:
I'd be happy to help you understand the concept of machine learning.

**What is Machine Learning?**

Machine learning is a way of teaching computers to make decisions or predictions on
their own, without being explicitly programmed. It's like teaching a computer to
recognize patterns in data, and then using that knowledge to make predictions or
decisions.

**How does Machine Learning work?**

Imagine you have a jar of different fruits, and you want to teach a computer to
recognize which ones are apples and which ones are bananas. You could write a list of
all the characteristics of each fruit, such as "apple: red, crunchy, sweet" and
"banana: yellow, soft, tastes like a string". The computer would then use that list to
recognize which fruits are apples and which are bananas.

In machine learning, we use algorithms to analyze data, identify patterns, and make
predictions. We can use this information to train a computer model to make decisions or
predictions on new, unseen data.
```

```
**Small Example:

============================
       AI TASK MENU
============================
1. Text Summarization
2. Sentiment Analysis
3. Interview Questions
4. AI Tutor
5. Exit

Enter your choice: 5

Exiting program...
```